Polytechnic Course Handbook

Basic Surveying Lab

Code: 2019 | Semester: II | Credits: 1.5 | Curriculum: REV2026

Hands-on experience with fundamental surveying instruments including theodolites, levels, and compasses for mapping and construction purposes.

Official Source Declaration

This content is strictly based on the **Official Syllabus PDF (Course Code: 2019)** provided by SITTTR Kerala for the Revision 2026 curriculum. All modules, experiments, and assessment schemes are extracted directly from the authoritative document.

Note: The assessment table in the PDF indicates a CIE total of 60 marks, derived from Attendance (5), Continuous Evaluation (30), Practical Tests (15), and Open-Ended Experiment (10).

Course Dashboard

Program	Civil Engineering and related branches
Course Type	Lab
Contact Hours	45 Hours / Semester

Teaching Scheme	0:0:3:0 (L:T:P:S)
CIE / ESE Marks	60 / 40
Pre-requisite	1002 - Fundamentals of Engineering Mathematics

How to Study This Course

1. Field Practice

Surveying is a practical skill. Focus on handling the instruments (leveling the dumpy level, centering the theodolite) during lab hours.

2. Record Maintenance

Maintain neat sketches and accurate calculations. 4 marks are specifically allocated for the Record in the ESE.

Course Outcomes (CO)

CO No.	Course Outcome	Bloom's Level
CO1	Plot land area using principles of chain survey and compass survey	Apply
CO2	Determine level difference between stations in the field	Apply
CO3	Prepare longitudinal profiles, cross profiles and contour maps using a leveling instrument	Apply
CO4	Measure the horizontal angles using theodolite	Apply

Syllabus Coverage

Module	Key Experiments	Hours
I	Intro to equipment, Area calculation, Compass survey	9
II	Simple Leveling, Rise & Fall, HI Method, Series Exam I	12
III	Profile Leveling, Fly Leveling, Contouring	9
IV	Repetition, Reiteration, Prolonging lines, Open-Ended Experiment	15

Learning Roadmap

The course begins with **Linear Measurements** (Tape/Chain) and **Angular Measurements** (Compass) in Module I. It then progresses to **Vertical Measurements** (Leveling) in Modules II & III, and finally introduces **Precision Angular Instruments** (Theodolite) in Module IV.

Module I: Linear & Compass Surveying

9 Instructional Hours | CO1

▼ Expt 1: Introduction to Surveying Equipment

Demonstration of basic equipment: Chain, Cross staff, Plane table, and Compass.

- **Chain/Tape:** For linear distances.
- **Cross Staff:** To set out perpendicular offsets.
- **Prismatic Compass:** To measure magnetic bearings of lines.

> **Malayalam support:** [illegible]

▼ Expt 2 & 3: Area Calculation & Compass Survey

Area Calculation: Determining the area of an open field using a tape by dividing the area into triangles.

Compass Survey: Determining inaccessible distances between points using a prismatic compass and tape.

Example: Inaccessible Distance

To find distance AB where B is across a river: Measure a baseline AC. Measure bearings of AB and CB. Use Sine Rule in triangle ABC.

Module II: Basic Leveling

12 Instructional Hours | CO2

▼ Expt 4, 5 & 6: Leveling Methods

Leveling is used to find the elevation (Reduced Level) of points relative to a datum.

1. Height of Instrument (HI) Method

Formula: $HI = RL_{BM} + BS$ and $RL_{Point} = HI - FS$ (or IS)

2. Rise and Fall Method

Calculates the difference between consecutive staff readings. A decrease in reading indicates a 'Rise', an increase indicates a 'Fall'.

HI Method Quick Calc

Benchmark RL (m): 100

Back Sight (BS) (m): 1.5

Fore Sight (FS) (m): 2.1

Module III: Advanced Leveling & Contouring

9 Instructional Hours | CO3

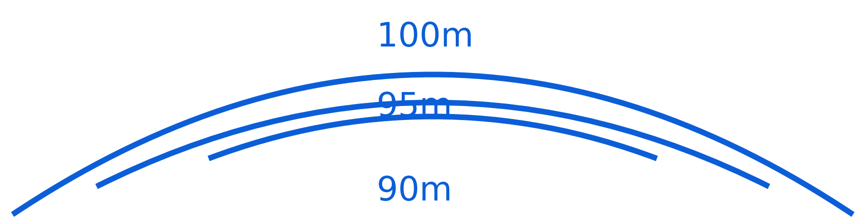
▼ Expt 7 & 8: Profile & Fly Leveling

Profile Leveling: Taking levels along a central line (Longitudinal) and across it (Cross-section) to determine the ground profile for roads or railways.

Fly Leveling: Conducted to check the accuracy of leveling or to transfer a benchmark to a distant point.

▼ Expt 9: Contouring

Preparation of a contour map by taking spot levels. Contours are lines joining points of equal elevation.



Conceptual Contour Lines for a Hill

Module IV: Theodolite Surveying

15 Instructional Hours | CO4

▼ Expt 10 & 11: Repetition & Reiteration

Methods to measure horizontal angles with high precision using a Transit Theodolite.

- **Repetition Method:** Measuring the same angle multiple times and taking the average to eliminate errors.
- **Reiteration Method:** Measuring several angles from a single station point.

> **Malayalam support:** [illegible] Repetition, Reiteration [illegible]. [illegible].

▼ Open Ended Experiment

Students must conceive and implement a project. Examples:

- Calculate pond capacity from a contour map.
- Measure area of a site with obstacles (buildings/trees).
- Sketch and calculate area of own house plot.

Student Activities for Self-Learning

Week	Activity Description	Hours
1-3	Study basic techniques, area determination using tape, compass surveying.	4.5
4-7	Identify Dumpy Level parts, study HI/Rise & Fall methods, Prep for Lab Exam 1.	6.0
8-10	Study Profile/Fly leveling and contour map preparation.	4.5
11-15	Theodolite adjustments, Repetition/Reiteration methods, Open-ended project prep.	7.5

Procedure & Formula Bank

HI Method

1. $HI = RL + BS$

2. $RL = HI - FS$ (or IS)

Check: $\sum BS - \sum FS = Last RL - First RL$

Rise & Fall Method

1. $Diff = Prev. Reading - Curr. Reading$

2. If +ve $\rightarrow$ Rise; If -ve $\rightarrow$ Fall

Check: $\sum BS - \sum FS = \sum Rise - \sum Fall = Last RL - First RL$

Visual Library

Contour Diagram

Refer to Lab Manual for:

- Prismatic Compass Parts
- Dumpy Level Cross-section
- Theodolite Vernier Scales

Assessment Methodology

Mode	Details	Marks
Attendance	End of semester	5
Continuous Evaluation (CE)	Every lab slot (50 marks scaled)	30
Practical Tests (CA1 & CA2)	7th & 14th week (40 marks each scaled)	15
Open-ended Experiment (OEE)	15th week (50 marks scaled)	10
Total CIE		**60**
ESE (Practical)	3-hour external exam	**40**

Module-wise Q&A

▼ **What is the difference between Back Sight (BS) and Fore Sight (FS)?**

Back Sight (BS): The first staff reading taken after the instrument is set up on a point of known elevation (Benchmark).

Fore Sight (FS): The last staff reading taken before shifting the instrument to another position.

▼ **Why is the Repetition method used in Theodolite surveying?**

It is used to measure horizontal angles with high precision. By measuring the angle multiple times and dividing the total by the number of repetitions, errors due to imperfect graduations and sighting are minimized.

▼ **What is a Contour Interval?**

The constant vertical distance between two consecutive contour lines is called the contour interval.

▼ **Explain 'Centering' in Theodolite.**

Centering is the process of placing the vertical axis of the theodolite exactly over the station mark using a plumb bob or optical plummet.

Practice Model Paper

(Not an official examination paper)

ESE Practical Exam (Sample)

Time: 3 Hours | Max Marks: 40

1. Write the procedure for determining the level difference between two points using the Rise and Fall method. (10 Marks)
2. Perform the experiment in the field and record the readings neatly in the tabular form. (10 Marks)
3. Calculate the Reduced Levels and apply arithmetic checks. (8 Marks)
4. Viva Voce and Record submission. (12 Marks)

Quick Revision

- **Chain Surveying:** Based on Triangulation.
- **Compass Surveying:** Based on Traversing.
- **Leveling:** BS is always on a known RL; FS is always the last point.
- **HI Method:** Faster for many Intermediate Sights (IS).
- **Rise & Fall:** More accurate as it provides a check on IS.
- **Theodolite:** Used for both horizontal and vertical angles.

Exam Tip: [illegible] [illegible] Arithmetic Check ($\sum BS - \sum FS$) [illegible] [illegible]. [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible].

Glossary

Benchmark (BM)
: A fixed point of known elevation above a datum.

Reduced Level (RL)
: The vertical distance of a point above or below the datum.

Parallax
: An apparent movement of the image relative to the cross-hairs, corrected by focusing.

Transiting
: The process of turning the theodolite telescope through 180° in a vertical plane.

References

- *Surveying I* by Punmia, B.C.; Jain, Ashok Kumar; Jain, Arun Kumar. Laxmi Publications.
- *Surveying and Levelling* by Basak, N. N. McGraw Hill Education.
- Web Resource: [NPTEL Surveying Course]

- Web Resource: Virtual Labs - Civil Engineering